

RARITONE – Backend Integration Guide

Overview

This is a 3-module AI forecasting system.

All modules depend on correct **dataset mapping**. Any mismatch in columns will break the system.

RARITONE consists of 3 independent forecasting modules using Facebook Prophet.

- Backend team MUST NOT change logic
- ONLY dataset + column mapping changes are allowed
- Every module depends on correct CSV structure alignment

1. GLOBAL DATA REQUIREMENTS

All datasets must include:

- Order Date → must be datetime
- Numeric column → Sales or Quantity
- No null values after cleaning

Required Cleaning

```
sales["Order Date"] = pd.to_datetime(sales["Order Date"], errors="coerce")
sales["Sales"] = pd.to_numeric(sales["Sales"], errors="coerce") # or Quantity
sales = sales.dropna(subset=["Order Date", "Sales"])
```

2. MODULE 1 — inventoryForecast.py

Purpose

Product-level forecasting + inventory planning

Required Columns(mandatory)

- Order Date
- Quantity
- Category/Sub-Category/Product (for filtering)

IF THEY ARE WITH DIFFERENT NAMES IN THE RARITONE DATASET CHANGE THE NAMES ACCORDING TO WHAT WE HAVE IN THE DATASET PROVIDED.

Changes Required

Dataset file:

CHANGE DATASET NAME FROM "Sample - Superstore.csv" TO RARITONE'S DATASET CSV FILE NAME.

CHANGE WHERE YOU SEE

```
sales = pd.read_csv("Sample - Superstore.csv")
```

Product filter:

CHANGE PRODUCT CATEGORY FROM "PHONE" TO WHICHEVER PRODUCT'S INVENTORY FORECAST YOU WANT TO SEE.

IF COLUMN NAME IS DIFFERENT "Sub-Category" TO WHATEVER COKUMN NAME THE DATASET HAS.

CHANGE WHERE YOU SEE:

```
product_sales = sales[sales["Sub-Category"] == "Phones"]
```

Change:

- Column name (Sub-Category/Category/Product)
- Product name (Phones/Laptop/etc.)
- Or remove filtering if not needed

Prophet format:

```
{"Order Date": "ds", "Quantity": "y"}
```

3. MODULE 2 — productDemand.py

Purpose

Overall demand forecasting (no filtering)

Required Columns(Mandatory)

- Order Date
- Sales

IF THEY ARE WITH DIFFERENT NAMES IN THE RARITONE DATASET CHANGE THE NAMES ACCORDING TO WHAT WE HAVE IN THE DATASET PROVIDED.

Changes Required

Dataset file:

CHANGE DATASET NAME FROM “Sample - Superstore.csv” TO RARITONE’S DATASET CSV FILE NAME.

CHANGE WHERE YOU SEE

```
sales = pd.read_csv("Sample - Superstore.csv")
```

Sales column mapping:

If “sales” is there use sales or else use different name used in raritone’s dataset like “revenue” or “amount”

CHANGE HERE

```
sales["Sales"] = pd.to_numeric(sales["Sales"], errors="coerce")
```

to

```
sales["Sales"] = pd.to_numeric(sales["Amount"], errors="coerce")
```

Prophet format:

```
{"Order Date": "ds", "Sales": "y"}
```

4. MODULE 3 — trendForecasting.py

Purpose

Trend + seasonality analysis

Required Columns

- Order Date
- Sales

Changes Required

Dataset file:

CHANGE DATASET NAME FROM “fashion_retail_sales.csv” TO RARITONE’S DATASET CSV FILE NAME.

CHANGE WHERE YOU SEE

```
sales = pd.read_csv("fashion_retail_sales.csv")
```

Sales column mapping:

If dataset uses different column names: SAME AS MODULE 2